PROJECT PROPOSAL & TECHNICAL REPORT

Garbage Classification using Image Processing and Computer Vision

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Course Information

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1. Project Proposal

Project Title

Smart Garbage Classification using Image Processing and Computer Vision

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Problem Definition

Manual garbage sorting is expensive, slow, and often inaccurate.

Different waste materials require different recycling and disposal methods. Incorrect classification reduces recycling efficiency and increases environmental damage.

The challenge is automatically identifying the type of waste from an image despite differences in:

- lighting
- background
- object shape
- object color
- image blur
- brightness variation

This makes image preprocessing a critical step before classification.

Project Objectives

This project focuses on building an intelligent waste classification system that can automatically recognize and classify garbage images into different categories such as plastic, glass, metal, paper, cardboard, biological waste, clothes, shoes, batteries, trash, and cartons.

The main objective of this project is to build a smart garbage classification system using Image Processing and Computer Vision techniques. Image Processing techniques are first applied to understand and enhance the dataset before moving to classification tasks.

These techniques include:

- image quality analysis
- histogram analysis
- brightness adjustment

- contrast adjustment
- image enhancement
- dataset balancing and preparation

The system aims to improve recycling efficiency and support smart waste management by reducing manual sorting errors and increasing automation in garbage classification.

Specific objectives include:

- Analyze and understand the image dataset
- Study image properties such as:
    - brightness
    - contrast
    - blur
    - histogram distribution
    - object visibility
- Apply image enhancement techniques including:
    - brightness adjustment
    - contrast adjustment
    - combined enhancement
- Improve image quality before model training
- Prepare a balanced dataset for future classification tasks
- Support automatic smart recycling systems using Computer Vision

This project is important because proper waste management is a major environmental challenge and Computer Vision can provide practical real-world solutions.

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2. Dataset Description

Dataset Name

Smart Waste Classification Dataset

Source

The dataset was collected from multiple public datasets available from Kaggle and other open-source image repositories.

Main sources include:

- Garbage Classification (12 classes) https://www.kaggle.com/datasets/mostafaabla/garbage-classification?select=garbage_classification
- Garbage Classification Dataset https://www.kaggle.com/datasets/manonstr/tipe-webscraping/data

The images were merged, cleaned, renamed, shuffled, and standardized to create a balanced and organized dataset for training and analysis.

Type

Natural Image Dataset

The dataset contains real-world images of waste materials captured in natural environments with different lighting conditions, object sizes, and backgrounds.

Why This Dataset Was Chosen

This dataset was selected because waste classification is an important real-world environmental problem.

Incorrect waste sorting causes:

- environmental pollution
- poor recycling efficiency
- increased landfill usage
- resource loss

This dataset is suitable because:

- it contains realistic images
- it has multiple important waste categories
- dataset size is sufficient for analysis
- it supports image processing techniques required in the course
- it provides strong practical application of Computer Vision

It also supports the development of smart recycling systems and sustainable environmental solutions.

Number of Images

Before balancing:

11,376 images

```
==================================================
DATASET SUMMARY
==================================================
glass        : 942 images
carton       : 1047 images
metal        : 1149 images
plastic      : 1205 images
trash        : 1140 images
shoes        : 988 images
battery      : 945 images
biological   : 985 images
cardboard    : 891 images
clothes      : 1034 images
paper        : 1050 images
==================================================
Total Images: 11376

Dataset is ready for train / validation / test split.
```

After balancing and train/validation/test split:

9,891 images

Number of Classes:

11 categories

- battery
- biological
- cardboard
- carton
- clothes
- glass
- metal
- paper
- plastic
- shoes
- trash

My practical balancing roadmap:

- **Balance classes to 900 each**
- **Train / Validation / Test split** (80 / 10 / 10)

```
============================================================
BALANCING + TRAIN / VAL / TEST SPLIT
============================================================
battery      → Train: 720 | Val: 90 | Test: 90
biological   → Train: 720 | Val: 90 | Test: 90
cardboard: only 891 images found
cardboard    → Train: 712 | Val: 89 | Test: 90
carton       → Train: 720 | Val: 90 | Test: 90
clothes      → Train: 720 | Val: 90 | Test: 90
glass        → Train: 720 | Val: 90 | Test: 90
metal        → Train: 720 | Val: 90 | Test: 90
paper        → Train: 720 | Val: 90 | Test: 90
plastic      → Train: 720 | Val: 90 | Test: 90
shoes        → Train: 720 | Val: 90 | Test: 90
trash        → Train: 720 | Val: 90 | Test: 90

============================================================
FINAL SUMMARY
============================================================
Train Total: 7912
Validation Total: 989
Test Total: 990
Grand Total: 9891
```

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3. Data Understanding - Sample Image Description

Each image contains a waste object or recyclable material such as:

- plastic bottles
- metal cans
- glass containers
- paper waste
- batteries
- clothes
- shoes
- household trash

The images vary in:

- size
- orientation
- brightness
- image quality
- background complexity
- camera angle

Some images are very clear while others include shadows, blur, or noisy backgrounds.

DATA UNDERSTANDING REPORT

Image Path: D:\ImgProject\split_dataset\train\plastic\plastic390.jpg

Type: RGB

Resolution: 474 x 631

Bit Depth: uint8

Brightness: 100.47 (Balanced)

Blur Score: 1095.10 (Clear)

Original Image

DATA UNDERSTANDING REPORT

Image Path: D:\ImgProject\split_dataset\train\plastic\plastic1.jpg

Type: RGB

Resolution: 270 x 186

Bit Depth: uint8

Brightness: 35.22 (Dark)

Blur Score: 484.60 (Clear)

Original Image

Image Type

The dataset images are RGB images (Red, Green, Blue), since they are captured from real-world environments using standard cameras and mobile devices. RGB format is important because object color is one of the main distinguishing features between waste categories such as plastic, glass, metal, and biological waste.

Some processing steps such as histogram analysis and blur detection were performed using grayscale conversion to simplify intensity analysis.

Resolution

Image resolution varies across the dataset because the images were collected from multiple public sources.

Common resolutions include:

- 270 × 186
- 512 × 384
- 640 × 480
- 1024 × 768

This variation is expected in real-world datasets and reflects practical image collection conditions.

To ensure consistency during future model training, all images will later be resized to a fixed input size.

Bit Depth

Most images use:

8-bit unsigned integer (uint8)

This means each pixel intensity ranges from:

0 → 255

which is the standard format for most image processing and computer vision applications.

This bit depth is sufficient for histogram analysis, brightness adjustment, contrast enhancement, and classification tasks.

Key Observations

Image Quality

Most images are clear and visually understandable, allowing the object to be recognized easily.

However, some images contain:

- low sharpness
- partial blur
- shadows

- complex backgrounds
- low lighting

These variations increase the difficulty of automatic classification and require preprocessing.

Blur detection using Variance of Laplacian showed that most training images are clear, while a smaller number contain moderate blur.

Lighting Variations

Lighting conditions vary significantly across the dataset.

Some images are:

- very dark
- overly bright
- balanced under natural lighting

Brightness analysis showed that dark images are common, especially in plastic and trash categories, where object visibility is sometimes reduced.

This affects feature extraction and model performance.

Brightness normalization helps improve visibility and consistency.

Presence of Noise or Blur

Some images contain background noise such as:

- multiple objects
- textured surfaces
- shadows
- reflections
- unrelated objects

Blur is also present in some samples due to:

- camera motion
- poor focus
- low image quality from original datasets

These issues make classification more difficult and justify the use of preprocessing techniques before training.

Object Characteristics

Objects vary significantly in:

- shape
- size
- color
- texture
- orientation

For example:

- plastic bottles may appear transparent or colored
- glass objects may reflect light strongly
- metal cans may have shiny surfaces
- clothes and shoes have irregular shapes
- biological waste has high visual diversity

This high variation makes classification challenging and requires robust image enhancement and feature extraction methods.

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4. Image Processing (Histogram + Brightness + Contrast Analysis)

Histogram Analysis

Histogram analysis was performed to study pixel intensity distribution and understand brightness and contrast levels inside the images.

The grayscale histogram represents how pixel values are distributed between dark (0) and bright (255) intensity levels.

This helps determine whether the image is:

- dark

- bright
- low contrast
- high contrast
- well balanced

Histogram Shape

Most sample images showed a left-skewed histogram, meaning a large number of pixels are concentrated in lower intensity values.

This indicates that many images are relatively dark.

Some images also showed wider histogram distribution, indicating better contrast and stronger visibility of object details.

Brightness Distribution

Brightness values varied across categories.

For example:

- plastic images often had lower brightness values
- paper and cardboard images often appeared brighter
- metal and glass images showed brightness variation due to reflections

Dark images produced concentrated histograms near the left side, while brighter images shifted toward the center or right.

Contrast Level

Low contrast images showed narrow histograms where intensity values were concentrated in a small range.

This reduced visibility of object edges and important features.

Higher contrast images showed wider histogram spread, improving detail recognition and separation between foreground and background.

Contrast enhancement helps improve classification quality.

Interpretation of Results

Histogram analysis confirmed that brightness inconsistency is one of the main challenges in the dataset.

Many images require preprocessing to improve:

- visibility
- contrast
- object clarity

This supports the need for brightness adjustment and contrast enhancement before classification.

Histogram analysis provides a strong foundation for image preprocessing decisions.

Visual Results for **train\plastic\plastic390.jpg:**

Original image before enhancement

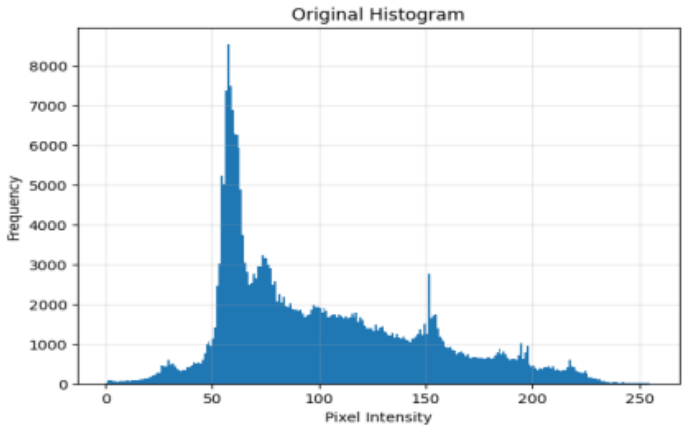


Image Brightness Value: 100.47

Before Processing

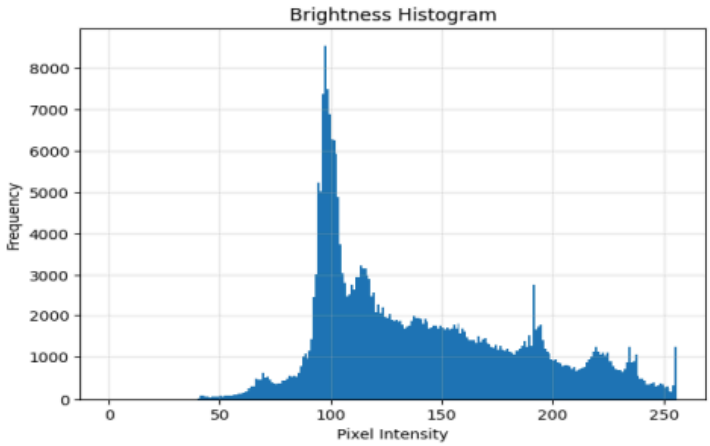
dark object visibility was weak

edges were unclear

low contrast reduced feature detection

some objects blended with the background

Brightness Adjustment:



Brightness Value After Adjustment: 139.78

Observation: Visibility improved by increasing brightness.

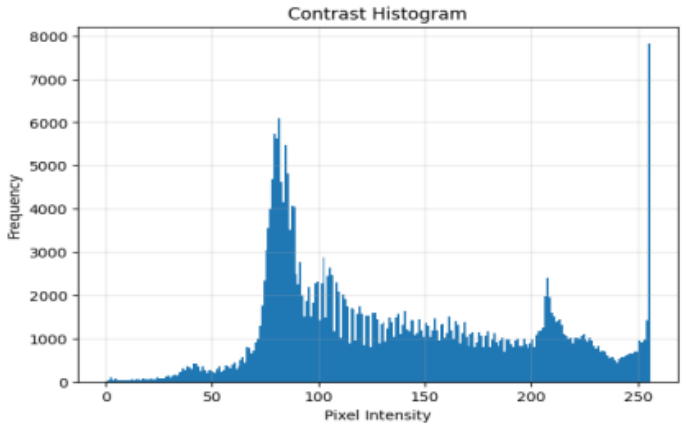
After Brightness Adjustment

object visibility improved

dark regions became clearer

important object details became easier to observe

Contrast Adjustment:



Contrast Result Brightness Value: 136.80

Observation: Contrast improved and object boundaries became clearer

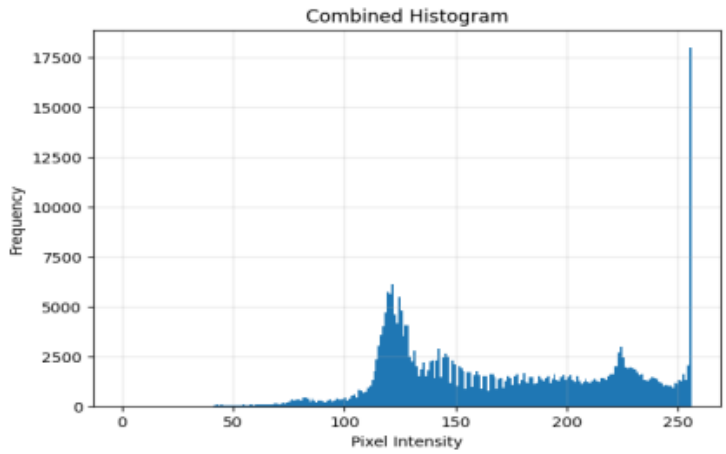
After Contrast Adjustment

object boundaries became sharper

texture details became more visible

separation between foreground and background improved

Brightness + Contrast Adjustments Combined:



Combined Adjustment Brightness Value: 170.95

Observation: Best enhancement result with clearer and brighter object appearance

After Combined Adjustment

both visibility and clarity improved

feature extraction conditions became stronger

the image became more suitable for classification tasks

Histogram Explanation

Histogram analysis provided a clear understanding of image intensity distribution.

Most original images showed left-skewed histograms, where pixel values were concentrated in lower intensity ranges. This indicates dark images with limited brightness.

After brightness adjustment, the histogram shifted toward the center and right side, showing improved illumination.

After contrast adjustment, the histogram became wider, indicating better intensity distribution and stronger separation between dark and bright regions.

Combined brightness and contrast adjustment produced the best results by improving both visibility and detail clarity.

This confirmed that histogram analysis is an effective method for deciding which preprocessing techniques should be applied.

5. Results & Insights

Summary of Key Findings

The analysis of the waste classification dataset showed that image preprocessing is an essential step before applying machine learning or deep learning models for classification.

The dataset contains a wide variety of real-world waste images collected under different environmental conditions. These variations include differences in:

- brightness
- contrast
- blur
- object size
- object orientation
- background complexity
- lighting conditions

Although most images are clear and usable, some images contain low brightness, poor contrast, shadows, reflections, and background noise, which can negatively affect classification performance.

Histogram analysis revealed that many images have pixel intensity distributions concentrated in darker regions, especially in categories such as plastic, trash, and biological waste. This indicates that brightness enhancement is necessary for better object visibility.

Contrast analysis also showed that several images have narrow intensity ranges, which reduces edge visibility and makes object boundaries less clear.

These findings confirm that preprocessing improves data quality and prepares the dataset for more accurate classification.

What Was Learned from the Analysis

Several important lessons were learned during dataset understanding and preprocessing.

First, image quality directly affects model performance. Even when the object is present clearly, poor lighting or blur can reduce feature extraction quality.

Second, histogram analysis is a powerful tool for understanding image characteristics. It helps identify whether an image is too dark, too bright, or has weak contrast before training begins.

Third, brightness and contrast adjustment significantly improve object visibility and feature clarity. Enhanced images make object shapes easier to detect and improve separation between the object and the background.

Fourth, dataset balancing is very important. Equalizing the number of images across classes helps prevent model bias toward larger categories and improves fairness during training.

This phase showed that successful classification begins with strong preprocessing, not only with model selection.

Important Patterns and Issues in the Dataset

Several patterns were observed during the analysis.

Common Patterns

- Plastic and trash images often appeared darker than other categories
- Glass and metal images frequently showed reflections and lighting variation
- Clothes and shoes had irregular object shapes and diverse textures
- Biological waste showed high variation in appearance and structure
- Paper and cardboard often had brighter images with simpler backgrounds

Common Issues

- background noise from unrelated objects
- shadows and reflections
- low brightness in indoor images
- inconsistent image resolution across sources

These issues increase classification difficulty and justify the use of image enhancement techniques before training.

Main Insights

The most important insight from this phase is that image preprocessing is not optional—it is necessary for building an effective classification system.

A strong model cannot perform well if the input images are poor in quality.

The project demonstrated that:

- preprocessing improves image quality
- histogram analysis helps guide enhancement decisions
- brightness and contrast adjustments improve feature visibility
- balanced datasets improve fairness and model stability
- real-world datasets require cleaning and standardization before training

The dataset is now properly prepared for the next phase, which is training classification models using machine learning and deep learning techniques.

This creates a strong foundation for building an intelligent smart garbage classification system that supports recycling efficiency and sustainable waste management.

Conclusion

This project focused on understanding and preparing a smart garbage classification dataset using Image Processing and Computer Vision techniques. The dataset was collected from multiple public sources, cleaned, standardized, balanced, and prepared for future classification tasks.

Through data understanding, histogram analysis, brightness adjustment, and contrast enhancement, several important challenges were identified, including lighting variation, blur, background noise, and inconsistent image quality. These issues can significantly affect classification accuracy if not handled properly.

The preprocessing phase showed that image enhancement improves object visibility, strengthens feature extraction, and creates better conditions for machine learning and deep learning models. Dataset balancing also helped ensure fair class distribution and reduced the risk of model bias.

This phase provided a strong foundation for the next stage of the project, which involves training and evaluating classification models for automatic waste recognition. The final goal is to support smart recycling systems and improve waste management using practical Computer Vision solutions.